



SCHOOL OF ARTIFICIAL INTELLIGENCE

Deep Learning

Syllabus

UDACITY.COM

Overview

Build foundational skills in deep learning by designing and training neural networks to solve complex real-world problems. You'll begin with the essentials of neural networks, advancing to specialized architectures like Convolutional and Recurrent Neural Networks, along with Generative Adversarial Networks. Through projects, create models for applications such as image classification, sentiment analysis, and face generation, gaining hands-on experience with PyTorch and advanced training techniques. Ideal for those aiming to harness the potential of deep learning, this experience prepares you to tackle AI challenges across various domains.

Nanodegree Program

Intermediate



61 hours



4.7 (978 Reviews)

Prerequisites

Prior to enrolling, you should have the following knowledge:

Pandas

Intermediate Python

Vectors

Basic calculus

NumPy

Deep learning fluency

Basic probability

Feedforward neural networks

Derivatives

Neural network basics

Python proficiency

Matrix operations

Matrix multiplication

Linear algebra

Fully convolutional neural networks

Differential calculus

matplotlib

Variational autoencoders

Jupyter notebooks

Machine learning frameworks in Python

You will also need to be able to communicate fluently and professionally in written and spoken English.

Skills You'll Learn

Introduction to Deep Learning

Neural networks | Deep learning | Gradient descent | Overfitting prevention | scikit-learn | Deep learning tools | Model performance metrics | Perceptron | PyTorch | Advanced probability | Training neural networks

Convolutional Neural Networks

Image pre-processing | Image segmentation | Neural network initialization | Bounding boxes | Semantic image segmentation | Convolutional kernels | Training neural networks | U-net | Convolutional neural networks | Autoencoders | Image classification | Transfer learning | Model training | Object detection | PyTorch

RNNs and Transformers

Training neural networks | Backpropagation | NLP transformers | Hyperparameter tuning | GPT | Recurrent neural networks | Bert | Long-short term memory networks | Neural networks | Word2vec



Building Generative Adversarial Networks

Generative adversarial networks | Model evaluation | Deep learning techniques | Markov games | Deep learning model optimization | Hyperparameter tuning | CycleGAN | Image generation

Courses

01. Welcome to the Deep Learning Nanodegree Program

45 minutes

The Deep Learning Nanodegree program offers a solid introduction to the world of artificial intelligence. In this program, you'll master fundamentals that will enable you to go further in the field, launch or advance a career, and join the next generation of deep learning talent that will help define a beneficial, new, AI-powered future for our world. You will study cutting-edge topics such as Neural Networks, Convolutional Neural Networks, Recurrent Neural Networks, Generative Adversarial Networks, and build projects in PyTorch.

- | | | |
|----|--|---|
| L1 | An Introduction to Your Nanodegree Program | Welcome! We're so glad you're here. Join us in learning a bit more about what to expect and ways to succeed. |
| L2 | Getting Help | You are starting a challenging but rewarding journey! Take 5 minutes to read how to get help with projects and content. |

02. Introduction to Deep Learning

20 hours

This course covers foundational deep learning theory and practice. We begin with how to think about deep learning and when it is the right tool to use. The course covers the fundamental algorithms of deep learning, deep learning architecture and goals, and interweaves the theory with implementation in PyTorch.

- | | | |
|----|---|---|
| L1 | Introduction to Deep Learning | Meet your instructor, get an overview of the course, and find a few interesting resources in this introductory lesson. |
| L2 | Deep Learning | This introductory lesson on Deep Learning covers how experts think about deep learning and how to know when deep learning is the right tool for the job, including some examples. |
| L3 | Minimizing Error Function with Gradient Descent | Beginning with PyTorch and moving into both Error Functions, Gradient Descent, and Backpropagation, this lesson provides an overview of foundational neural network concepts. |
| L4 | Intro to Neural Networks | This introduction to neural networks explains how algorithms inspired by the human brain operate and puts to use those concepts when designing a neural network to solve particular problems. |
| L5 | Training Neural Networks | Learn how to train neural networks and avoid overfitting or underfitting by employing techniques like Early Stopping, Regularization, Dropout, Local Minima, and Random Restart! |



L6

Project:

Developing a Handwritten Digits Classifier with PyTorch

In this project, you will use your skills in designing and training neural networks to classify handwritten digits using the well-known MNIST dataset.

03. Convolutional Neural Networks

21 hours

This course introduces Convolutional Neural Networks, the most widely used type of neural networks specialized in image processing. You will learn the main characteristics of CNNs that make them so useful for image processing, their inner workings, and how to build them from scratch to complete image classification tasks. You will learn what are the most successful CNN architectures, and what are their main characteristics. You will apply these architectures to custom datasets using transfer learning. You will also learn about autoencoders, a very important architecture at the basis of many modern CNNs, and how to use them for anomaly detection as well as image denoising. Finally, you will learn how to use CNNs for object detection and semantic segmentation.

L1

Introduction to CNNs

In this lesson we will look at the main applications of CNNs, understand professional roles involved in the development of a CNN-based application, and learn about the history of CNNs.

L2

CNN Concepts

In this lesson we will recap how to use a Multi-Layer Perceptron for image classification, understand the limitations of this approach, and learn how CNNs can overcome these limitations.

L3

CNNs in Depth

In this lesson we will study in depth the basic layers used in CNNs, build a CNN from scratch in PyTorch, use it to classify images, improve its performance, and export it for production.

L4

Transfer Learning

In this lesson we will learn about key CNN architectures and their innovations, and apply multiple ways of adapting them to our use cases with transfer learning.

L5

Autoencoders

In this lesson we will design and train linear and CNN-based autoencoders for anomaly detection and for image denoising.

L6

Object Detection and Segmentation

In this lesson we will study applications of CNNs beyond image classification. We will train and evaluate an object detection model as well as a semantic segmentation model on custom datasets.

L7

Project:

Landmark Classification & Tagging for Social Media

In this project, you will apply the skills you have acquired in the Convolutional Neural Network (CNN) course to build a landmark classifier.

04. RNNs and Transformers

3 hours

This course covers multiple RNN architectures and discusses design patterns for those models. You'll also learn about transformer architectures.

- L1 Intro to RNN
- L2 Introduction to LSTM
- L3 Introduction to Transformers
- L4 **Project:**
Text Translation and Sentiment
Analysis using Transformers

05. Building Generative Adversarial Networks

15 hours

Learn to understand and implement a Deep Convolutional GAN (generative adversarial network) to generate realistic images, with Ian Goodfellow, the inventor of GANs, and Jun-Yan Zhu, the creator of CycleGANs.

- L1 Introduction to Generative Adversarial Networks Introduction to this course, prerequisites, and your course instructor.
- L2 Generative Adversarial Networks Ian Goodfellow, the inventor of GANs, introduces you to these exciting models. You'll also implement your own GAN on the MNIST dataset.
- L3 Training a Deep Convolutional GANs In this lesson, you'll implement a Deep Convolution GAN to generate complex color images.
- L4 Image to Image Translation Jun-Yan Zhu, one of the creators of the CycleGAN, will lead you through Pix2Pix and CycleGAN formulations that learn to do image-to-image translation tasks.
- L5 Modern GANs In this lesson, you will implement more advanced GAN architectural techniques that have had a significant impact on the realism of generated images.
- L6 **Project:**
Face Generation Define two adversarial networks, a generator, and a discriminator, and train them until you can generate realistic faces.

06. Congratulations!

10 minutes

Congratulations on finishing your program!

L1 Congratulations!

Congratulations on your graduation from this program! Please join us in celebrating your accomplishments.

Meet Your Instructors



Giacomo Vianello

Director, Machine Learning Engineer

Giacomo Vianello is Director, Machine Learning Engineer at Moody's, where he leads AI innovation and deploys ML solutions across insurance and real estate. He's passionate about demystifying advanced ML concepts and teaching engineers how to build production-grade AI systems that actually ship.



Nathan Klarer

CEO

Nathan is an expert in the scientific fields of bioengineering-bioinformatics and machine learning. He is responsible for building multiple successful enterprises that have driven tens of millions of dollars of investment globally.



Erick Galinkin

Principal AI Researcher

Erick Galinkin is a hacker and computer scientist, leading research at the intersection of security and artificial intelligence at Rapid7. He has spoken at numerous industry and academic conferences on topics ranging from malware development to game theory in security.



Thomas Hossler

Sr Deep Learning Engineer

Thomas is originally a geophysicist but his passion for Computer Vision led him to become a Deep Learning engineer at various startups. By creating online courses, he is hoping to make education more accessible. When he is not coding, Thomas can be found in the mountains skiing or climbing.

Why Udacity



Demonstrate proficiency with practical projects

Projects are based on real-world scenarios and challenges, allowing you to apply the skills you learn to practical situations, while giving you real hands-on experience

- ✓ Gain proven experience
- ✓ Retain knowledge longer
- ✓ Apply new skills immediately



24/7 access to real human support

Reviewers provide timely and constructive feedback on your project submissions, highlighting areas of improvement and offering practical tips to enhance your work

- ✓ Get help from subject matter experts
- ✓ Gain valuable insights and improve your skills
- ✓ Learn industry best practices